

Daniel Waxman

Basis Research Institute
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EDUCATION

- Ph.D. Electrical Engineering, *December 2025*
 Stony Brook University
 Thesis: *Sequential Bayesian Learning and Ensembles for Online Inference and Experimental Design*
 Advisor: Petar M. Djurić
- B.S. Mathematics & Applied Mathematics and Statistics (*cum laude*), May 2021
 Stony Brook University

EMPLOYMENT & ACADEMIC APPOINTMENTS

- 01/26 – Postdoctoral Research Scientist
Present *Basis Research Institute*, Cambridge, MA.
 Postdoctoral fellowship jointly supervised by Matthew Levine (Basis) and Youssef Marzouk (MIT).
- 05/26 – Research Affiliate
Present *Laboratory for Information & Decision Systems*
 Massachusetts Institute of Technology, Cambridge, MA.
- 06/25 – Research Intern
01/26 *Basis Research Institute*, New York, NY.
 Research in causal inference, probabilistic programming, and applied Bayesian machine learning with Rafal Urbaniak and Jack Feser.

RESEARCH INTERESTS

- Bayesian Machine Learning — particularly problems in dynamical systems and online settings
- Ensembling Methods — particularly Bayesian-based ensembling rules and online ensembling
- Causal Inference — particularly causal discovery and causality for dynamical systems

PUBLICATIONS

* denotes equal contribution

Journal Papers

- 2026 M. Levine, **D. Waxman**, I. Urteaga. “CD-Dynamax: A JAX-based Python package for continuous-discrete probabilistic state space modeling and inference.” *Submitted*
- 2026 K. Lamer, E. V. Davis, Z. Mages, Z. Zhu, N. E. Rojas-Robles, K. N. Mitu, E. R. Vivoni, Z. Wang, H. Li, M. Cadetdu, P. Joshi, **D. Waxman**. “The Encanto Neighborhood Field Experiments Dataset: Land–Atmosphere Interactions in Urban Phoenix.” *Submitted*.

- 2026 **D. Waxman**, F. Llorente, P. M. Djurić, “Sequential Inference with Gaussian Processes: A Signal Processing Perspective.” Accepted to *IEEE Signal Processing Magazine*.
[arXiv]
- 2026 **D. Waxman**, F. Llorente, P. M. Djurić. “Bayesian Ensembling: Insights from Online Optimization and Empirical Bayes.” *Transactions on Machine Learning Research (TMLR)*, 2026.
[OpenReview] [arXiv] [code]
- 2025 **D. Waxman**, F. Llorente, K. Lamer, P. M. Djurić. “Designing an Optimal Sensor Network via Minimizing Information Loss.” *Bayesian Analysis*, 2025.
[ProjectEuclid] [arXiv] [code]
- 2024 **D. Waxman**, P. M. Djurić. “Dynamic Online Ensembles of Basis Expansions.” *Transactions on Machine Learning Research (TMLR)*, 2024.
[OpenReview] [arXiv] [code]
- 2024 **D. Waxman**, K. Butler, P. M. Djurić. “DAGMA-DCE: Interpretable, Non-Parametric Differentiable Causal Discovery.” *IEEE Open Journal of Signal Processing*, vol. 5, pp. 393-401, 2024.
[IEEE Xplore] [arXiv] [code]

Conference Papers

- 2026 M. Heinonen, Y. Shenfeld, R. Baptista, **D. Waxman**, D. Batenkov, T. Cooijmans, E. Bingham. “Wasserstein Residuals: Learning Gradient Flows from Population Dynamics.” Accepted to *ICML 2026 Workshop on Structured Probabilistic Inference & Generative Modeling (SPIGM)*.
- 2026 F. Llorente, **D. Waxman**, S. Jantre, N. Urban, S. Minkoff. “Robust, Online, and Adaptive Decentralized Gaussian Processes.” *Proceedings of the 2026 IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, 2026.
[IEEEXplore] [arXiv]
- 2025 **D. Waxman**, P. M. Djurić. “Non-Stationary Causal Learning via Hierarchical Modeling.” *Proceedings of the 2025 Asilomar Conference on Signals, Systems, and Computers*, 2025.
[IEEEXplore]
- 2025 F. Llorente*, **D. Waxman***, P. M. Djurić. “Decentralized Online Ensembles of Gaussian Processes for Multi-Agent Systems.” In *Proceedings of the 2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2025.
[IEEEXplore] [arXiv] [code]
- 2024 K. Butler*, **D. Waxman***, P. M. Djurić. “Tangent Space Causal Inference: Leveraging Vector Fields for Causal Discovery in Dynamical Systems.” In *Advances in Neural Information Processing Systems*, 37, 120078-120102, 2024.
[OpenReview] [arXiv] [code]
- 2024 **D. Waxman**, P. M. Djurić. “A Gaussian Process-based Streaming Algorithm for Prediction of Time Series With Regimes and Outliers.” In *Proceedings of the 27th International Conference on Information Fusion (FUSION)*, Venice, Italy, 2024.
[IEEEXplore] [arXiv] [code]
- 2023 M. Ajirak, **D. Waxman**, F. Llorente, P. M. Djurić. “Fusion of Gaussian Process Predictions with Monte Carlo Sampling.” *Proceedings of the 57th Asilomar Conference on Signals, Systems, and Computers* (pp. 1367-1371), 2023.
[IEEE Xplore] [arXiv]
- 2023 Y. Liu, C. Cui, **D. Waxman**, K. Butler, P. M. Djurić. “Detecting Confounders in Multivariate Time Series Using Strength of Causation.” In *Proceedings of the 31st European Signal Processing Conference (EUSIPCO)*, pp. 1400-1404, 2023.
[IEEE Xplore] [pdf] [video]

Professional Meeting Abstracts

- 2026 K. Lamer, E. V. Davis, Z. J. Mages, P. Joshi, N. Rojas, E. R. Vivoni, Z. Wang, Z. Zhu, **D. Waxman**. “Multisensor Observations and Meter-Scale Modeling of Urban Park Cooling Dynamics for Digital Twin Integration.” *106th AMS Annual Meeting*.
- 2024 D. Chernoff, **D. Waxman**, G. Brite, L. Langman, C Rabeno, L. Czerwonka, P. M. Djurić. “The Development of an Automated Algorithm to Identify and Manage Post-thyroidectomy Hypocalcemia.” *American Head and Neck Society 2024 General Meeting*.

RESEARCH TALKS

- 2026 ISBA World Meeting ’26
Session on “Calibrated Bayes: Model Design and Adaptation Under Limited Resources”
Talk Title: “Online & Predictively-Oriented Model Fusion”
- 2026 SIAM Uncertainty Quantification ’26
Minisymposium on “Sequential Inference for Dynamical Systems”
Talk Title: “GP4SP: Sequential Gaussian Process Inference for Signal Processing”
- 2026 Queens College Computer Science Colloquium
Talk Title: “Online Bayesian Learning and Ensembles” [\[slides\]](#)
- 2024 2024 SIAM New York-New Jersey-Pennsylvania Section Conference
Talk Title: “Optimizing Observation Locations via Minimizing Information Loss”
- 2023 Acoustics Research Institute of the Austrian Academy of Sciences
Talk Title: “Causal Discovery via Quantifying Influences” [\[link\]](#) [\[slides\]](#)
- 2023 Bellairs Workshop on Machine Learning and Statistical Signal Processing for Data on Graphs
Talk Title: “Bayesian Combination”

TEACHING

As Instructor of Record

- Fall 2025 Understanding Machine Learning (ESE 188 @ SBU)
Spr 2025 Understanding Machine Learning (ESE 188 @ SBU)

As Teaching Assistant

- Sum 2022 Random Signals and Systems (ESE 306 @ SBU)
Spr 2022 Random Signals and Systems (ESE 306 @ SBU)
Fall 2021 Programming Fundamentals (ESE 124 @ SBU)

OUTREACH & SERVICE

Reviewing

I’ve served as a reviewer for 60+ manuscripts across top-tier journals and conferences in machine learning and signal processing, including several top reviewer awards.

- Journals IEEE Transactions on Signal Processing [2026]
 IEEE Transactions on Aerospace and Electronic Systems [2026]
 Transactions on Machine Learning Research (TMLR) [2024, 2025, 2026]
 Journal of Open Source Software (JOSS) [2024, 2025, 2026]
 EURASIP Journal on Advances in Signal Processing [2023]
- Conf. NeurIPS [2025; **Top Reviewer** 2025]
 ICLR [2025, 2026]
 ICML [2025, **Position Track Gold Reviewer** 2026]
 European Signal Processing Conference (EUSIPCO) [2024, 2025, 2026]
 NeurIPS Workshops [2024]
 IEEE Statistical Signal Processing Workshop [2025]
 Asilomar Conferences on Signals, Systems, and Computers (ASSC) [2025, 2026]
 AAAI [2026]

University Service

- 2021–25 Stony Brook University Graduate Student Organization
 Primary Senator, Dept. of Electrical & Computer Engineering
 Graduate Student Representative for Graduate Council
 Reviewer for Distinguished Travel Award
- 2022–23 SBU Strategic Planning Committee [\[link\]](#)
 Member of Strategic Planning Committee; helped draft vision statement for Project REACH

Expository Talks

- 2024 SBU ECE Honors Seminar
 Ensembling and Fusion: Perspectives and Solutions from Signal Processing
- 2022 Stony Brook Math Club
 Making Markov Chains with Metropolis [\[video\]](#)

Organizing

- 2026 Special Session on “Bayesian Learning and Inference on Graphs”
 Co-organized with Kurt Butler, Victor Elvira, and Petar M. Djurić
 Asilomar 2026, Pacific Grove, CA
- 2026 Minisymposium on “Sequential Inference for Dynamical Systems”
 Co-organized with Fernando Llorente-Fernandez
 SIAM UQ 2026, Minneapolis, MN.
- 2025 Special Session on “Advances in Causal Inference: Theory and Applications”
 Co-organized with Petar M. Djurić
 Asilomar 2025, Pacific Grove, CA

Mentoring

- 2022–25 CUNY Directed Reading Program
 Mike Prebil (*Gaussian Processes and Bayesian Optimization*)
 Edgar Cuapio Diaz (*Gaussian Processes and Bayesian Optimization*)
 Jonathan Jaimangal (*Hamiltonian Monte Carlo*, **Winner Outstanding Poster Award**)
 Masroor Khonkhodzaev (*Dynamic Programming and Reinforcement Learning*)
 Percy Martinez (*Fourier Analysis*)
 Isabella Chittumuri (*Elements of Statistical Learning*)

2023 Stony Brook DRP
 Shailen Smith (*Probabilistic Machine Learning*)

2022 Bayesian Data Analysis for the Global South
 Volunteer TA for online course by Aki Vehtari [[certificate](#)]

Updated May 2026